

Homework (1)

665 Course

Problem 2.3r

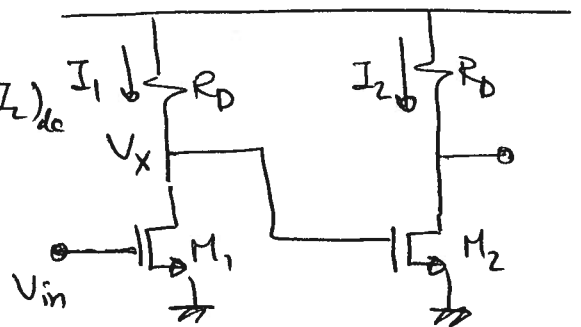
- Identical CS amplifiers $\rightarrow$ same bias, same size.
- 2nd order non-linearity only per stage.
- Using ideal square law behavior (saturation mode).
- $\rightarrow$ find IIP3 of the cascaded amplifier.

assume: $(U_{in})_{dc} = (V_{out})_{dc} = (U_x)_{dc}$

$$= V_{dd} - R_D (I_1)_{dc} = V_{dd} - R_D (I_2)_{dc}$$

$$I_1 = \frac{k_n}{2} \left(\frac{W}{L}\right)_n (V_{gs1} - V_{th})^2$$

$$I_2 = \frac{k_n}{2} \left(\frac{W}{L}\right)_n (V_{gs2} - V_{th})^2$$



Cascade of CS stage

$$I_1 = \frac{k_n}{2} \left(\frac{W}{L}\right)_n (V_{in} - V_{th})^2, \quad I_2 = \frac{k_n}{2} \left(\frac{W}{L}\right)_n (U_x - V_{th})^2$$

$$(I_1)_{dc} = (I_2)_{dc} = \frac{k_n}{2} \left(\frac{W}{L}\right)_n ((U_x)_{dc} - V_{th})^2 = \frac{k_n}{2} \left(\frac{W}{L}\right)_n V_{dsat}^2$$

$$V_{dsat1} = V_{dsat2} = V_{dd} - I_{1,dc} R_D - V_{th} = V_{dd} - \frac{k_n}{2} \left(\frac{W}{L}\right)_n V_{dsat}^2 R_D - V_{th}$$

$$i_1 = \frac{k_n}{2} \left(\frac{W}{L}\right)_n [V_{dsat} + A \sin(\omega t)]^2 \quad \text{since } U_{in} = (U_{in})_{dc} + A \sin(\omega t)$$

$$U_x = V_{dd} - i_1 R_D = V_{dd} - \frac{k_n}{2} \left(\frac{W}{L}\right)_n R_D [V_{dsat}^2 + 2 V_{dsat} A \sin(\omega t) + A^2 \sin^2(\omega t)]$$

$$V_x = V_{dsat} + V_{tn} + \underbrace{\frac{K_n}{2} \left(\frac{W}{L}\right)_n R_D V_{dsat}}_{g_m R_D} \cdot 2 \cdot A \sin(\omega t) - \frac{K_n}{2} \left(\frac{W}{L}\right)_n R_D A^2 \sin^2$$

$$i_2 = \frac{K_n}{2} \left(\frac{W}{L}\right)_n \left[V_x - V_{tn} \right]^2$$

$$= \frac{K_n}{2} \left(\frac{W}{L}\right)_n \left[V_{dsat} - \underbrace{\frac{K_n}{2} \left(\frac{W}{L}\right)_n R_D A^2 \sin^2(\omega t)}_{x^2} - g_m R_D \underbrace{A \sin(\omega t)}_x \right]^2$$

$$i_2 = \frac{K_n}{2} \left(\frac{W}{L}\right)_n \left[V_{dsat} - \frac{K_n}{2} \left(\frac{W}{L}\right)_n R_D x^2 - g_m R_D x \right]^2$$

$$= \frac{K_n}{2} \left(\frac{W}{L}\right)_n \left[V_{dsat}^2 - 2 V_{dsat} \left[g_m R_D x + \frac{K_n}{2} \left(\frac{W}{L}\right)_n R_D x^2 \right] \right.$$

$$\left. + \left[g_m^2 R_D^2 x^2 + 2 g_m R_D x \cdot \frac{K_n}{2} \left(\frac{W}{L}\right)_n R_D x^2 - \dots \right] \right]$$

$$i_2 \approx \frac{K_n}{2} \left(\frac{W}{L}\right)_n V_{dsat}^2 - g_m^2 R_D x - g_m \frac{K_n}{2} \left(\frac{W}{L}\right)_n R_D x^2 + g_m^2 R_D^2 x^2 \frac{K_n}{2} \left(\frac{W}{L}\right)_n$$

$$+ \left[\frac{K_n}{2} \left(\frac{W}{L}\right)_n R_D \right]^2 2 g_m x^3$$

$$i_2 \approx I_{2,dc} - g_m^2 R_D x - \frac{g_m^2 R_D}{2 V_{dsat}} x^2 + \frac{g_m^3 R_D^2}{2 V_{dsat}} x^2 + \frac{g_m^3 R_D^2}{2 V_{dsat}^2} x^3$$

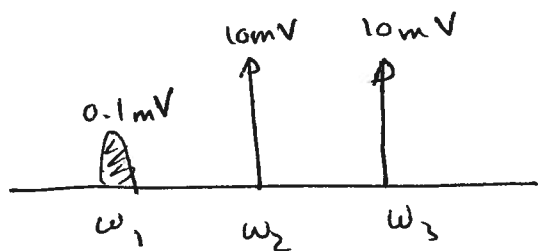
$$V_{out} \approx V_{dd} - i_2 R_D \approx V_{dd} - I_{2,dc} R_D + g_m^2 R_D^2 x$$

$$+ \frac{g_m^2 R_D^2}{2 V_{dsat}} x^2 - \frac{g_m^3 R_D^3}{2 V_{dsat}} x^2 + \frac{g_m^3 R_D^3}{2 V_{dsat}^2} x^3$$

$$\alpha_1 = g_m^2 R_D^2, \quad \alpha_2 = g_m^2 R_D^2 \left(\frac{1}{2 V_{dsat}} - \frac{g_m R_D}{2 V_{dsat}} \right), \quad \alpha_3 = - \frac{g_m^3 R_D^3}{2 V_{dsat}^2}$$

$$A_{HP3} = \sqrt{\left| \frac{4 \alpha_1}{3 \alpha_3} \right|} = \sqrt{\frac{4 g_m^2 R_D^2 \times 2 V_{dsat}^2}{3 g_m^3 R_D^3}} = \sqrt{\frac{8}{3} \frac{1}{g_m R_D}} V_{dsat}$$

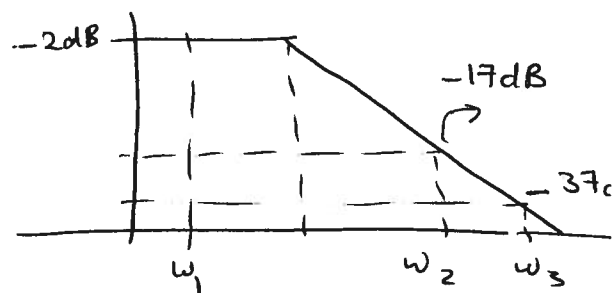
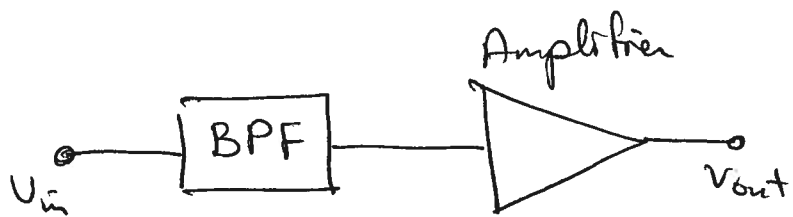
Problem 2.5r



$$\omega_3 - \omega_2 = \omega_2 - \omega_1$$

$$\Rightarrow \boxed{2\omega_2 - \omega_3 = \omega_1} \Rightarrow \text{IM3 product}$$

will have same frequency as desired.

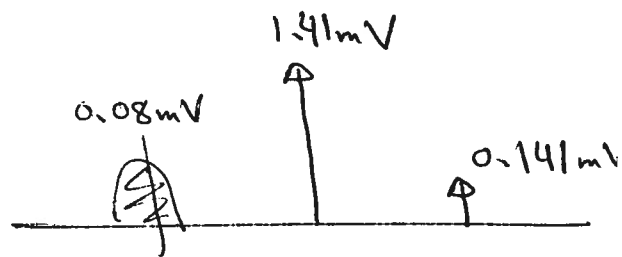


- (a) Calculate IIP3 of amplifier, such that IM3 product falling @ ω_1 is 20 dB lower than the desired signal.

gain of BPF @ $\omega_1 = -2 \text{ dB} \rightarrow 0.794$

@ $\omega_2 = -17 \text{ dB} \rightarrow 0.141$

@ $\omega_3 = -37 \text{ dB} \rightarrow 0.0141$



$$y(t) = \alpha_1 x(t) + \alpha_2 x^2(t) + \alpha_3 x^3(t)$$

$$x(t) = \underbrace{A_1 \sin(\omega_1 t)}_{\text{neglected}} + A_2 \sin(\omega_2 t) + A_3 \sin(\omega_3 t)$$

$$y(t) = \alpha_1 [A_2 \sin(\omega_2 t) + A_3 \sin(\omega_3 t)] + \alpha_3 [A_2^3 \sin^3(\omega_2 t) + 3A_2^2 A_3 \sin^2(\omega_2 t) \sin(\omega_3 t) + 3A_2 A_3^2 \sin(\omega_2 t) \sin^2(\omega_3 t) + A_3^3 \sin^3(\omega_3 t)]$$

$$\begin{aligned}
 \sin^2(\omega_2 t) \sin(\omega_3 t) &= \frac{1}{2} [1 - \cos(2\omega_2 t)] \sin(\omega_3 t) \\
 &= \frac{1}{2} \left[\sin(\omega_3 t) - \frac{1}{2} \sin(\omega_3 + 2\omega_2)t - \frac{1}{2} \sin(\omega_3 - 2\omega_2)t \right] \\
 &= \frac{1}{2} \sin(\omega_3 t) + \frac{1}{4} \sin(2\omega_2 - \omega_3)t - \frac{1}{4} \sin(2\omega_2 + \omega_3)t
 \end{aligned}$$

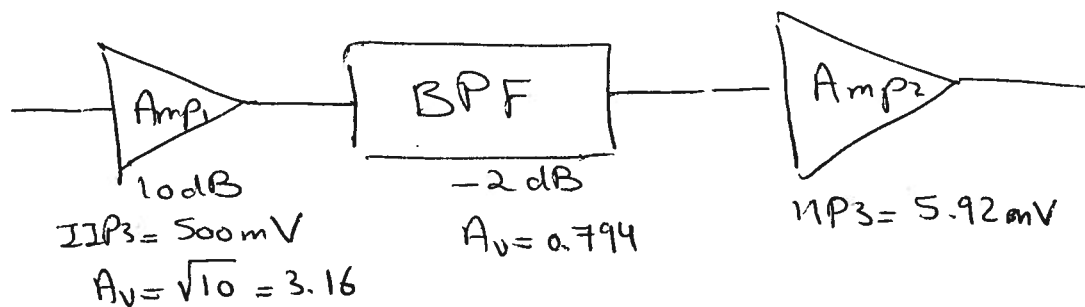
$$y(t) = \alpha_1 [A_2 \sin(\omega_2 t) + A_3 \sin(\omega_3 t)] + \alpha_3 \frac{3A_2^2 A_3}{4} \sin(2\omega_2 - \omega_3)t$$

$$\alpha_1 A_1 = \frac{3\alpha_3}{4} \times A_2^2 A_3 \times 10 \rightarrow 20 \text{ dB factor.}$$

$$= \frac{3\alpha_3}{4\alpha_1} = \frac{A_1}{10 A_2^2 A_3} \rightarrow A_{IP3} = \sqrt{\frac{4}{3} \left| \frac{\alpha_3}{\alpha_1} \right|} = \sqrt{\frac{10 A_2^2 A_3}{A_1}}$$

$$A_{IP3} = A_2 \sqrt{\frac{10 A_3}{A_1}} = 1.41 \sqrt{\frac{1.41}{0.08}} = 5.92 \text{ mV}$$

(b) If a voltage gain of 10 dB & IIP3 = 500 mV.



first amplifier is very linear and has voltage gain of 3.16

⇒ linearity is dominated by Amp₂. (second stage).

$$IIP3 = \frac{5.92 \text{ mV}}{\sqrt{10} \times 0.8} \approx 2.36 \text{ V} \Rightarrow \text{in band IIP3.}$$

out of band IIP3 ⇒

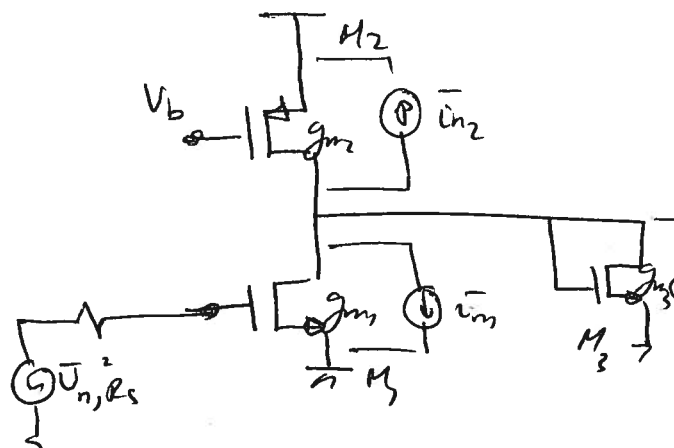
$$\begin{aligned}
 IIP3(\omega_2) &= 13.26 \text{ mV} \\
 IIP3(\omega_3) &= 128.3 \text{ mV}
 \end{aligned}$$

Problem 2.21.

Determine NF, neglect channel length modulation (r_o) & body effect (g_{mb}).

(c) Consider only drain noise of MOSFET.

- each MOSFET device has a parallel noise current.



$$\bar{U}_{n,out}^2 = \bar{U}_{n,Rs}^2 \left(\frac{g_{m1}}{g_{m3}} \right)^2 + \left[\bar{i}_{n1}^2 + \bar{i}_{n2}^2 + \bar{i}_{n3}^2 \right] \left(\frac{1}{g_{m3}} \right)^2.$$

$$= 4KTR_s \left(\frac{g_{m1}}{g_{m3}} \right)^2 + 4KT\gamma \left[g_{m1} + g_{m2} + g_{m3} \right] \left(\frac{1}{g_{m3}} \right)^2.$$

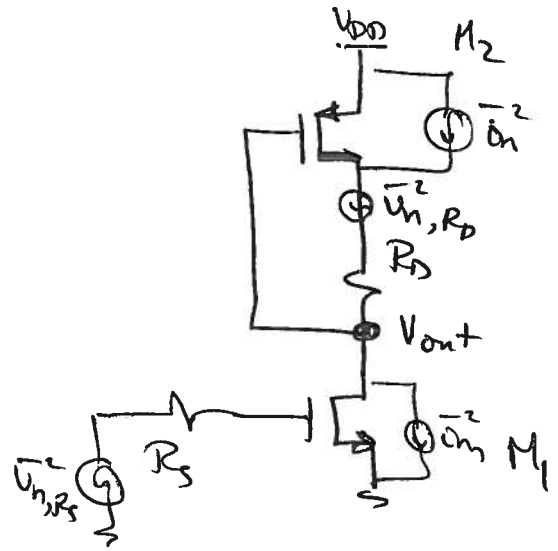
$$NF = \frac{\bar{U}_{n,out}^2}{\bar{U}_{n,out}^2 \text{ due to } R_s} = 1 + \frac{\gamma}{R_s} \frac{g_{m1} + g_{m2} + g_{m3}}{g_{m1}^2}.$$

$$= 1 + \frac{\gamma}{g_{m1} R_s} + \frac{\gamma}{g_{m1} R_s} \left(\frac{g_{m2}}{g_{m1}} + \frac{g_{m3}}{g_{m1}} \right).$$

(e)

- The load in this circuit can be considered as simple diode connected device.

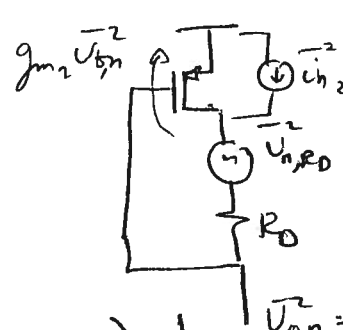
- R_D will not affect the load impedance at dc (no capacitance).



or the noise of the load

$$\bar{U}_{o,n}^2 = \bar{U}_{n,RS}^2 \left[\frac{g_{m1}}{g_{m2}} \right]^2 + (\bar{i}_{n1}^2 + \bar{i}_{n2}^2) \frac{1}{g_{m2}^2}$$

$$= 4KT R_S \left(\frac{g_{m1}}{g_{m2}} \right)^2 + 4KT \gamma (g_{m1} + g_{m2}) \frac{1}{g_{m2}^2}$$

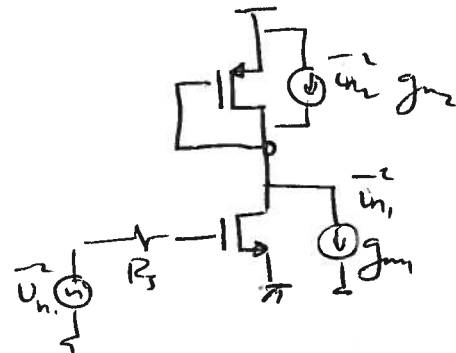


$$\bar{U}_{o,n}^2 = \frac{\bar{i}_n^2}{g_{m2}^2} \quad Z_{in} = \frac{1}{g_{m2}}$$

$$NF = \frac{\bar{U}_{o,n}^2}{\bar{U}_{o,n}^2 (\text{due to } R_S)} = 1 + \frac{\gamma (g_{m1} + g_{m2})}{R_S g_{m2}^2}$$

$$= 1 + \frac{\gamma}{g_{m1} R_S} + \frac{\gamma}{g_{m1} R_S} \frac{g_{m2}}{g_{m1}}$$

$$= 1 + \frac{\gamma}{g_{m1} R_S} \left(1 + \frac{g_{m2}}{g_{m1}} \right)$$



14 Problem 2.23: (b) neglect r_o & g_{mb} .

When there is loop in the circuit,
it is preferable to use
superposition studying the effect
of each noise source alone.

- $\bar{i}_{n,R_s}^2$ & $\bar{i}_{n,M_2}^2$ are similar.

$$\bar{i}_{n,x}^2 = \bar{i}_{n,R_s}^2 + \bar{i}_{n,x}^2$$

KCL @ $V_{x,n}$ noise due to $\bar{i}_{n,x}$.

$$\frac{V_x}{R_s} + g_{m1} V_x + g_{m2} V_{o,n} = \bar{i}_{n,x}$$

$$V_{o,n} = g_{m1} V_x R_D$$

$$V_x \left[\frac{1}{R_s} + g_{m1} + g_{m1} g_{m2} R_D \right] = \bar{i}_{n,x}$$

$$\bar{V}_{o,n} = \frac{\bar{i}_{n,x}}{1/R_s + g_{m1} (1 + g_{m2} R_D)} \cdot g_{m1} R_D = \frac{\bar{i}_n \cdot R_D}{1 + g_{m1} R_s (1 + g_{m2} R_D)} g_{m1} R_s$$

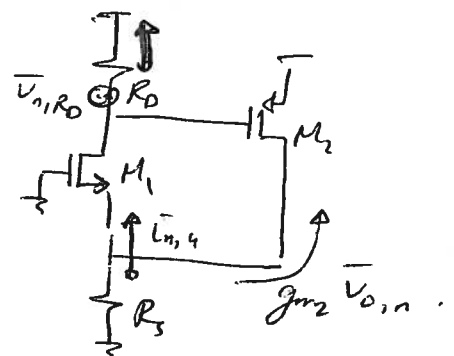
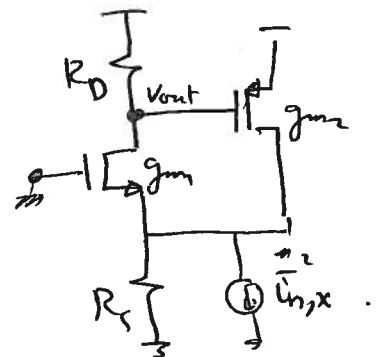
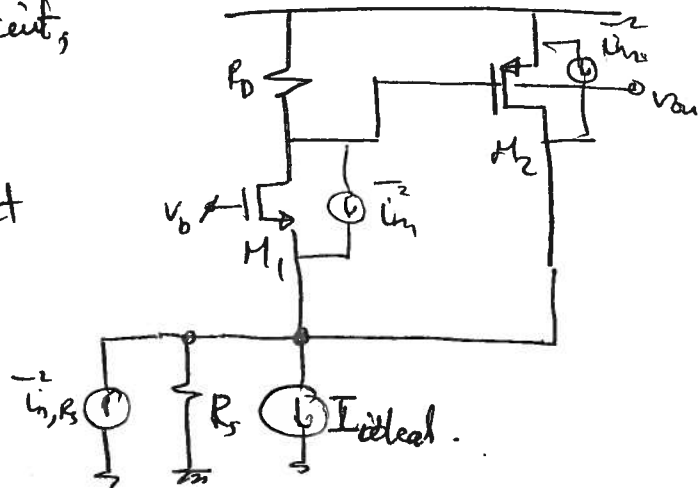
Noise due to R_D :

$$\bar{V}_{o,n} = \bar{V}_{n,R_D} + \bar{i}_{n,q} R_D$$

$$\bar{i}_{n,q} = -g_{m1} [0 + (\bar{i}_{n,q} + g_{m2} \bar{V}_{o,n})] R_s$$

$$\bar{i}_{n,q} = \frac{-g_{m1} g_{m2} R_s}{1 + g_{m1} R_s} \bar{V}_{o,n}$$

$$\bar{V}_{o,n} = \bar{V}_{n,R_D} - \frac{g_{m1} g_{m2} R_s R_D}{1 + g_{m1} R_s} \bar{V}_{o,n}$$



$$\overline{U_{o,n}} \approx \frac{1 + g_{m1} R_s}{1 + g_{m1} R_s (1 + g_{m2} R_D)} \overline{U_{n,RD}}$$

Noise due to M_{1r}

$$\overline{U_{o,n}} = - \overline{i_{n,s}} R_D$$

$$\overline{U_{x,n}} = (\overline{i_{n,s}} + g_{m2} R_D \overline{i_{n,s}}) R_s$$

$$\therefore \overline{U_{x,n}} = \overline{i_{n,s}} \frac{R_s (1 + g_{m2} R_D)}{1}$$

$$\overline{i_{n,s}} = \overline{i_{n,1}} - g_{m1} \overline{U_{x,n}}$$

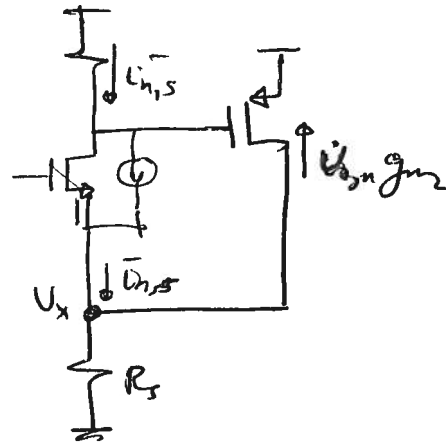
$$= \overline{i_{n,1}} - g_{m1} R_s (1 + g_{m2} R_D) \overline{i_{n,s}}$$

$$\therefore \overline{i_{n,s}} = \frac{\overline{i_{n,1}}}{1 + g_{m1} R_s (1 + g_{m2} R_D)}$$

$$\overline{U_{o,n}} = \frac{-\overline{i_{n,1}} R_D}{1 + g_{m1} R_s (1 + g_{m2} R_D)}$$

$$NF = 1 + \gamma g_{m2} R_s + \frac{\gamma g_{m1} R_s}{g_{m1}^2 R_s^2} + \frac{4KT R_D}{4KT R_D^2 / R_s} \left(\frac{1 + g_{m1} R_s}{g_{m1} R_s} \right)$$

$$X/F = 1 + \gamma g_{m2} R_s + \frac{\gamma}{g_{m1} R_s} + \frac{R_s}{R_D} \left(1 + \frac{1}{g_{m1} R_s} \right)^2$$



Problem 2.23 (C)

- noise of M_1 & M_2 are parallel current sources.

→ Same analysis applied.

R_S noise

$$\bar{U}_{n,x} = \bar{U}_{n,R_S} - (g_{m1} + g_{m2}) R_S \bar{U}_{x,n}$$

$$\bar{U}_{x,n} = \frac{\bar{U}_{n,R_S}}{1 + \underbrace{(g_{m1} + g_{m2}) R_S}_{g_m}}$$

$$\bar{U}_{o,n} = \bar{U}_{x,n} - (g_m) R_F \bar{U}_{x,n} = \bar{U}_{x,n} [1 - g_m R_F] = \bar{U}_{n,R_S} \left(\frac{1 - g_m R_F}{1 + g_m R_S} \right)$$

M_1 & M_2 noise

$\bar{i}_{n3}$ models noise from M_1 & M_2

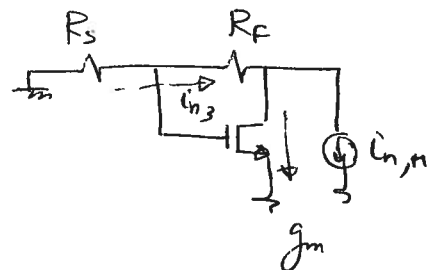
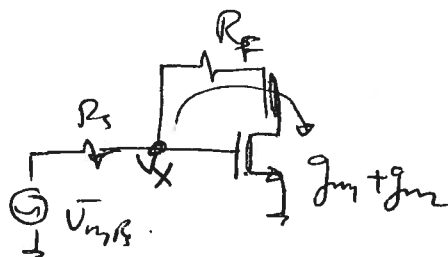
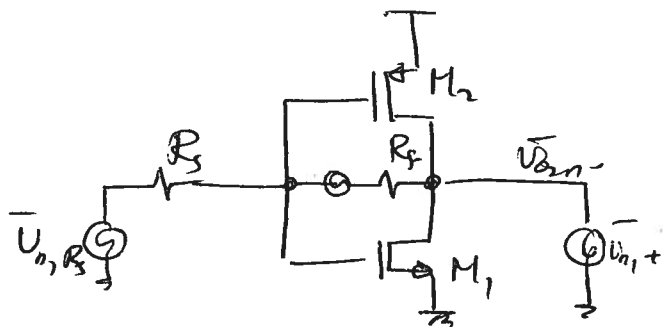
$$\bar{i}_{n3} = 4KT (g_{m1} + g_{m2}) = 4KT \gamma g_m$$

$$\bar{U}_{n,x} = -\bar{i}_{n3} R_S, \quad \bar{U}_{o,n} = -\bar{i}_{n3} (R_S + R_F)$$

$$\bar{i}_n = \bar{i}_{n3} - g_m \bar{U}_{n,x} = \bar{i}_{n3} + g_m R_S \bar{i}_{n3}$$

$$\bar{i}_n = \bar{i}_{n3} (1 + g_m R_S) \longrightarrow \bar{i}_{n3} = \frac{\bar{i}_n}{1 + g_m R_S}$$

$$\bar{U}_{o,n} = - \frac{(R_S + R_F)}{(1 + g_m R_S)} \bar{i}_n$$



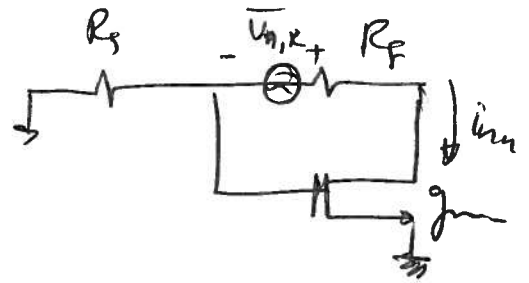
noise due to R_F

$$\bar{U}_{n,x} = -\bar{I}_{n_4} R_S$$

$$\bar{U}_{o,n} = \bar{U}_{n,x} + \bar{U}_{n,R_F} - \bar{I}_{n_4} R_F$$

$$\bar{U}_{n_4} = g_m \bar{U}_{n,x} = -g_m R_S \bar{I}_{n_4} \longrightarrow \bar{I}_{n_4} = 0$$

$$\bar{U}_{o,n} = \bar{U}_{n,R_F} = \sqrt{4KT R_F}$$



$$NF = \frac{\bar{U}_{o,n} \text{ (total)}}{\bar{U}_{o,n} \text{ (due to } R_S)}$$

$$NF = 1 + \underbrace{\frac{R_F}{R_S} \frac{(1+g_m R_S)^2}{(1-g_m R_F)^2}}_{\text{noise due to } R_D} + \underbrace{\frac{\gamma g_m (R_S + R_F)^2}{R_S (1-g_m R_F)^2}}_{\text{noise due to } M_1 \& M_2}$$

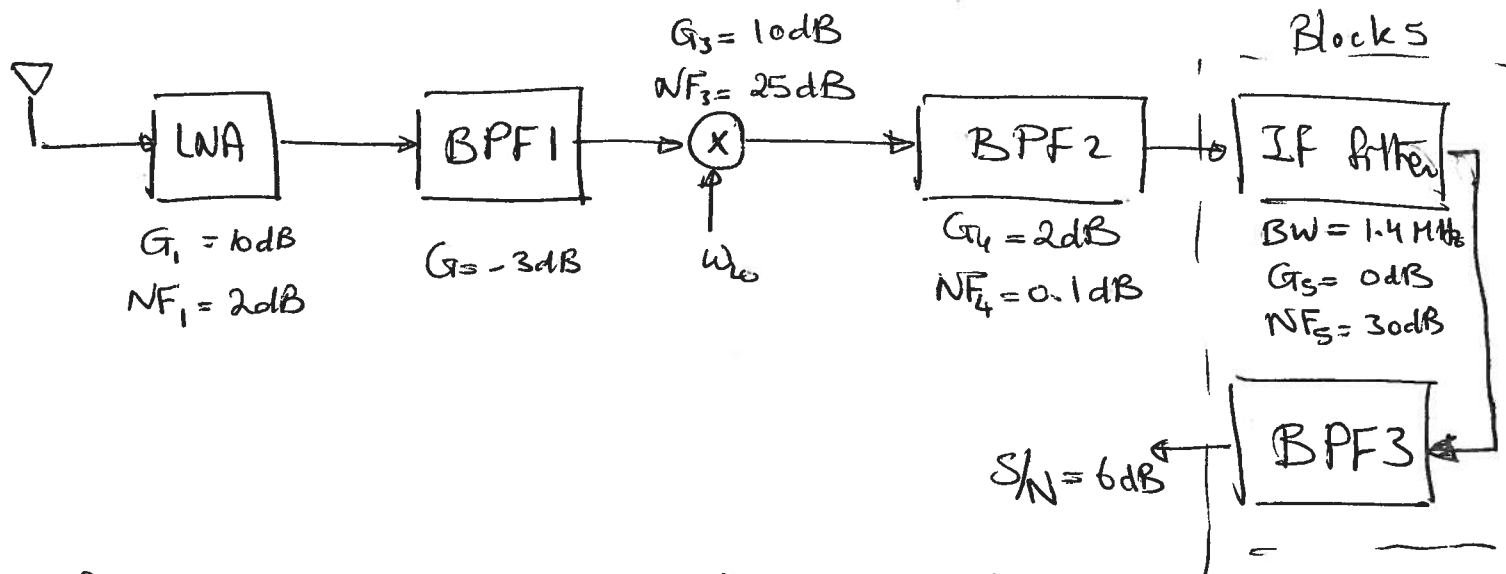
naturally $g_m R_F \gg 1$:

$$NF = 1 + \frac{R_F}{R_S} \left(\frac{g_m R_S + 1}{g_m R_F - 1} \right)^2 + \frac{\gamma}{g_m R_S} \frac{g_m^2 (R_F + R_S)^2}{(g_m R_F - 1)^2}$$

for matching: $g_m R_S \approx 1$ & $g_m R_F \gg 1$ high gain Amp

$$NF = 1 + \gamma \left(1 + \frac{R_S}{R_F} \right)^2 + \frac{4}{g_m R_F}$$

Problem 5r



→ Assume all blocks are all matched to 50Ω.

$$\rightarrow NF_{ht} = NF_1 + \frac{NF_2 - 1}{G_1} + \frac{NF_3 - 1}{G_1 G_2} + \frac{NF_4 - 1}{G_1 G_2 G_3} + \frac{NF_5 - 1}{G_1 G_2 G_3 G_4}$$

$$NF_{2dB} = -G_{dB} = 3dB \left\{ \text{Assume BPF1, } NF = \text{loss} \right\}.$$

$$NF_{ht} = 1.585 + \frac{2-1}{10} + \frac{316-1}{10 \times 1/2} + \frac{1.023-1}{10 \times \frac{1}{2} \times 10} + \frac{1000-1}{10 \times \frac{1}{2} \times 10 \times 1.585}$$

$$= 1.585 + 0.1 + 63 + 4.6 \times 10^{-4} + 12.61 = 77.3$$

$$NF_{htdB} = 10 \log_{10}(77.3) = 18.88dB$$

$NF_{dB} = 18.88dB$

$$\text{Total Sensitivity} = SNR_0 + NF_{dB} + \underbrace{(-174 \text{ dBm/Hz} + 10 \log BW)}_{\text{noise power}}$$

$$= 6 + 18.88 + (-174 + 10 \log(1.4)) = -87.7 \text{ dBm}$$

Total IIP3 of RX is -15 dBm .

SFDR = (Max. power where IM3 = noise) - Sensitivity.

$$= \frac{2 P_{IIP3} + (-174 \text{ dBm} + NF + 10 \log B)}{3} - \text{Sensitivity.}$$

$$= -41.2 \text{ dBm} - 87.7 \text{ dBm} = 46.5 \text{ dB}$$

$$= \frac{2 (P_{IIP3} + 174 \text{ dBm} - NF - 10 \log B)}{3} - SNR_{\min}$$

$$\boxed{\text{SFDR} = 46.5 \text{ dB}}$$

